MOBIN ROZATI

Tehran, Iran • +98 912 899 4900 • thisizzmobinrt@gmail.com • LinkedIn • GitHub

Backend Software Engineer focused on Python, Django, FastAPI, scalable backend architecture, and AI-enabled systems. Experience spans event-driven backend design, database optimization, LLM/MCP integrations, and applied machine learning/computer vision using PyTorch and OpenCV.

EDUCATION

B.Sc. Computer Engineering | K.N. Toosi University of Technology (KNTU) *Sept 2022 - Present*

- **Relevant Coursework:** Computer Vision (19.5/20), Artificial Intelligence (18.25/20)
- **Research Focus:** Multi-Agent Deep Reinforcement Learning (DRL) and Digital Twin (DT)-enabled Vehicular Edge Computing.

TECHNICAL SKILLS

Languages	Java, Python, SQL, C, English (Fluent)
Backend Web	FastAPI, Django, Django REST Framework, RESTful APIs, Celery, WebSockets, ASGI
Databases & Search	PostgreSQL, MySQL, Redis, MongoDB, Oracle Database
Architecture	OOP, Domain-Driven Design (DDD), Clean Architecture, SOLID, JWT, Microservices, Multithreading, Event-Driven Architecture, Transactional Outbox
AI & Data	Agentic AI, LLM Integration, RAG, LangChain, LangGraph, Web Crawling (BeautifulSoup, Playwright), Prompt Engineering
Computer Vision & ML	PyTorch, OpenCV, CNNs, Image Processing, Image Restoration, Edge/Corner Detection, Homography & Perspective Geometry, SIFT/Feature Matching, RANSAC, Image Registration, Synthetic Data Generation, OCR Evaluation
Testing	Unit Testing, Integration Testing, Compatibility Testing
Tools & DevOps	Git, Docker, Linux (Fedora, Ubuntu), Postman

WORK EXPERIENCE

Cryptocurrency Trading Simulation Platform

Backend Software Engineer

- Led the architectural refactoring of a Django-based cryptocurrency trading engine, addressing singleton execution, database hot paths, Redis coupling, and concurrency bottlenecks.
- Designed a deterministic, event-driven architecture with account-based execution sharding, durable event processing, idempotency, replay, and crash recovery.
- Developed a behavior-preserving migration strategy and compatibility tests for order execution, positions, margin, PnL, liquidation, drawdown rules, and real-time market-data processing.

Backend Software Engineer | Hozhan Company *June 2024 - March 2025*

- Engineered a secure, high-performance B2B backend architecture, resolving critical performance bottlenecks while ensuring strict backward compatibility with older legacy client technologies.
- Designed and optimized relational databases using PostgreSQL and MySQL, ensuring highly efficient data retrieval and synchronization across distributed services.
- Collaborated with cross-functional teams to integrate APIs and deploy containerized applications using Docker and Linux environments.

WORK EXPERIENCE (CONTINUED)

AI Software Engineering Intern | AI & Software Internship
Jul 2025 - Sep 2025

- Spearheaded the design and implementation of a FastMCP Server Integration project during an intensive 3-month summer internship.
- Connected an advanced MCP server with Large Language Models (LLMs), facilitating the seamless sharing of tools like RAG pipelines and external APIs.
- Enhanced the model's contextual retrieval capabilities, laying the foundation for scalable, AI-driven backend services.

PROJECTS & RESEARCH

Document Scanning & Enhancement [Project Link]

Python, PyTorch, OpenCV, CNNs, Homography, Computer Vision

- Built an end-to-end document-scanning pipeline combining four-corner heatmap localization, perspective rectification, and residual encoder-decoder enhancement.
- Designed a paired synthetic-data pipeline with perspective transforms, backgrounds, illumination/shadow variation, blur, noise, JPEG degradation, and source-level train/validation/test separation.
- Achieved 29.35 dB PSNR / 0.934 canonical SSIM for synthetic enhancement and 0.53 px mean heatmap corner error with 99.5% all-corners success; evaluated real-world OCR/domain-gap failure modes and hardened the project with 173 automated tests.

High-Performance CSV Data Pipeline [Project Link]

Java, MySQL, OOP

- Designed and implemented a memory-efficient backend utility engineered to stream and parse massive datasets concurrently without triggering out-of-memory errors.
- Developed a configuration engine enabling dynamic, user-defined schema rendering and conditional runtime relational-table compilation.
- Built a dual-stream transactional validation pipeline to isolate semantic parsing errors and log corrupt payloads directly into a structured error matrix without interrupting execution flow.

AI-Powered Web Application (Mini GPT) [Project Link]

FastAPI, Hugging Face, Python

- Developed a seamless real-time communication platform integrating advanced open-source Language Models via Hugging Face.
- Implemented non-blocking REST APIs with FastAPI to handle concurrent user requests efficiently.

University Management System [Project Link]

FastAPI, Clean Architecture

- Architected a backend system for university operations applying Domain-Driven Design (DDD) and Clean Architecture principles, drastically improving code maintainability.

Divar Platform Simulation [Project Link]

Java, OOP

- Built a localized simulation of a classified ads platform using Java, emphasizing robust Object-Oriented Programming and efficient data parsing.